TARGET Protein

The effect of augmented administration of enteral protein to critically ill adults on clinical outcomes: A cluster randomised, cross-sectional double cross-over, registry-embedded, pragmatic clinical trial

Chief Investigator:

Associate Professor Adam Deane
Deputy Director ICU, Head of ICU Research
Intensive Care, The Royal Melbourne Hospital- City Campus
Grattan Street, Parkville Victoria 3050
Phone: (03) 9342 9253
Email: adam.deane@mh.org.au

Project Manager and Site Principal Investigator:

Dr Lee-anne Chapple
Senior Critical Care Dietitian, Intensive Care, Royal Adelaide Hospital
Research Fellow, University of Adelaide
Phone: (08) 7074 1763
Email: lee-anne.chapple@adelaide.edu.au

Coordination Centre:

Intensive Care Research
Royal Adelaide Hospital
Level 4, 4G751
Port Road
Adelaide, South Australia, 5000
AUSTRALIA
Phone: (08) 7074 1765
Email: targetprotein@adelaide.edu.au

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1. MANAGEMENT COMMITTEE

Professor Rinaldo Bellomo
MBBS, MD, FRACP, FCICM, FAAHMS
Professor of Intensive Care, The University of Melbourne
Co-director, ANZ Intensive Care Research Centre
Director of Intensive Care Research, Austin Hospital
Email: rinaldo.bellomo@austin.org.au

Professor Marianne Chapman
MBBS, PhD, FJFICM, FANZCA
Director of Research, Intensive Care Unit
Royal Adelaide Hospital
E-mail: marianne.chapman@sa.gov.au

Dr Lee-anne Chapple
BMedSci, MNutDiet, PhD
Post-Doctoral Research Fellow, Dietitian
Adelaide Medical School, University of Adelaide
Intensive Care Research, Royal Adelaide Hospital
E-mail: lee-anne.chapple@adelaide.edu.au

Associate Professor Adam Deane
MBBS, PhD, FRACP, FCICM
Deputy Director ICU, Head of ICU Research
Intensive Care, The Royal Melbourne Hospital
Email: adam.deane@mh.org.au

Dr Suzie Ferrie
GradDipFoodTech, BAPhil, BSc, MND, PhD
Critical Care Dietitian
Intensive Care, Royal Prince Alfred Hospital
Email: suzie.ferrie@health.nsw.gov.au

Dr Mark Finnis
MBBS, FANZCA, FCICM, MBiostat
Statistician

Australian and New Zealand Intensive Care Research Centre, Monash University
Email: mark.finnis@adelaide.edu.au

Professor Craig French
MBBS, FANZCA, FFICANZCA, FCICM
Director of Intensive Care
Sunshine Hospital, Western Health
Email: Craig.French@wh.org.au

Ms Sally Hurford
Post Grad Dip Clinical Research
ICU Clinical Trials Project Manager
Medical Research Institute of New Zealand
Email: sally.hurford@ccdhb.org.nz

Dr Nima Kakho
MBBS, FACEM, FCICM
Intensive Care Staff Specialist
University Hospital Geelong, Barwon Health
Email: nima.kakho@barwonhealth.org.au

Dr Emily Karahalios
BSc (Hons), MPH, PhD
Senior Lecturer in Biostatistics, University of Melbourne
Email: emily.karahalios@unimelb.edu.au

Dr Matthew Maiden
BSc, MBBS, PhD, DRANZCOG, FJFICM, FACEM
Intensive Care Staff Specialist
Intensive Care, The Royal Melbourne Hospital
Email: mjmaiden@ozemail.com.au

Ms Stephanie O'Connor
RN, Grad. Dip. Card. Nsg, MNSc.
Clinical Services Coordinator/ ICU Research Manager
Royal Adelaide Hospital

E-mail: Stephanie.O’Connor@sa.gov.au

Professor Sandra Peake
BMBS, BSc(Hons), PHD, FCICM
Senior ICU Clinician
Intensive Care, The Queen Elizabeth Hospital
Email: Sandra.Peake@sa.gov.au

Associate Professor Jeffrey Presneill
MBBS, PhD, MBiostat, PGDipEcho, FRACP, FCICM
Intensivist and Deputy Director and Head of Safety and Quality
Intensive Care, The Royal Melbourne Hospital
Email: jeffrey.presneill@mh.org.au

Dr Emma Ridley
BNutDiet, MPH PhD
Senior Research Fellow
Australian and New Zealand Intensive Care Research Centre
Monash University
Email: emma.ridley@monash.edu

Mr Matthew Summers
BSc, MDiet
Project Manager / Research Dietitian
Intensive Care Research, Royal Adelaide Hospital
E-mail: matthew.summers@sa.gov.au

Dr An Duy Tran
PhD
Senior Research Fellow
Health Economics Unit, University of Melbourne
E-mail: an.tran@unimelb.edu.au

Ms Tricia Williams
RGN, BNP (Int Care)
Research Coordinator

Intensive Care, The Queen Elizabeth Hospital
Email: tricia.williams@sa.gov.au

Professor Paul Young
MB ChB BSc (Hons), FCICM, PhD
Senior ICU Clinician
Department of Critical Care
Wellington Regional Hospital, New Zealand
Email: paul.young@ccdhb.org.nz

Associate Investigator to PRO-SCAN sub-study
Dr Minh-Son To
BPharm, BMaCompSc, MBBS, PhD
Radiology Registrar
South Australia Medical Imaging, Central Adelaide Local Health Network
Email: Minh-Son.To@sa.gov.au

2. STUDY SITES AND SITE CONTACT

Coordinating Centre:
Royal Adelaide Hospital

Study Sites:
Austin Hospital – Melbourne, Victoria
PI: Professor Rinaldo Bellomo

Sunshine Hospital - Melbourne, Victoria
PI: Professor Craig French

Royal Adelaide Hospital – Adelaide, South Australia
PI: Dr Lee-anne Chapple

Royal Melbourne Hospital - Melbourne, Victoria
PI: Associate Professor Adam Deane

The Queen Elizabeth Hospital - Adelaide, South Australia

PI: Professor Sandra Peake

<u>The Royal Prince Alfred Hospital – Sydney, New South Wales</u>
PI: Dr Suzie Ferrie

<u>University Hospital Geelong - Geelong, Victoria</u>
PI: Dr Nima Kakho

<u>Wellington Regional Hospital – New Zealand</u>
PI: Professor Paul Young

3. LAY SUMMARY

An intensive care unit (ICU) stay is associated with significant muscle wasting in up to 80% of critically ill patients. This muscle wasting results in 'ICU-acquired weakness' that is associated with slower weaning from ventilator support, longer time to discharge alive from ICU and hospital, higher in-hospital costs which persist well after discharge from the acute care setting. Nutrition therapy, delivered as liquid nutrition via a tube into the stomach, forms usual care for critically ill patients. The provision of additional protein has the potential to attenuate muscle atrophy, and hence improve outcomes following critical illness. Current international guidelines recommend delivery of protein prescription of 1.2 - 2.0 g/kg/day or higher, but this is based on very low quality of evidence, and there is observational evidence to support both benefit and harm of augmented protein prescription. Therefore, there is a need for a high-quality randomised controlled trial of differing protein prescriptions. We propose a cluster randomised clinical trial with the aim of evaluating the effect of higher dietary protein delivery on outcomes of critically ill adult patients when compared to usual care. In this study each ICU will be randomised to one of the two study formulae for 3 months and then crosses over to the other study formulae for 3 months, which is then repeated, with each site participating for 12 months. Each eligible patient admitted to the ICU during these 3-month periods will receive the same study formula.

4. BACKGROUND

Each year around 130,000 Australians are admitted to an ICU and > 40,000 patients have an ICU admission that requires longer (> 24 hours) organ support and receive artificial enteral nutrition[1]. The mortality rate of this patient population is ~25%[2]. Rapid and substantial muscle wasting occurs in up to 80% of critically ill patients[3]. This muscle wasting causes 'ICU-acquired weakness', which is associated acutely with increased mortality, slower weaning from ventilator support, longer time to discharge alive from ICU and hospital, and higher in-hospital costs[4]. The detrimental effects of ICU-acquired weakness also persist long after discharge from the acute care setting and adversely affect return to work and quality of life in ICU survivors[5].

Our group has published preliminary evidence from a single-centre randomised clinical trial that increasing dietary protein during ICU admission reduces muscle wasting (augmented protein vs. usual protein: attenuation of quadriceps muscle layer thickness loss of 0.22 cm (95% CI, 0.06 to 0.38; P=0.01)[6]. Accordingly, there is a plausible mechanistic pathway that augmenting dietary protein will attenuate muscle loss and reduce ICU-acquired weakness, the duration of mechanical ventilation, and complications of and duration of hospitalisation. We also have preliminary evidence that augmenting dietary protein has a meaningful impact on outcomes that are important to patients.

Observational data in favour of augmenting protein delivery

Prospective and retrospective observational studies have reported that augmenting enteral protein (protein delivered at ≥ 1.2 g/kg/day) is associated with reduced mortality, shorter time to discharge alive, reduced duration of mechanical ventilation, and reduced duration of ICU and hospital admission[7-12]. However, the impact of unmeasured confounders in such observational data may be substantial. Nonetheless, based solely on observational data, international critical care nutrition guidelines recommend delivery of dietary protein of ≥ 1.2 g/kg body weight/day[13, 14]. Due to the poor quality of available evidence these recommendations are graded as 'consensus only' or 'very low quality'.

Observational data against augmenting protein delivery

Whilst most observational studies suggest benefit with more protein, there are also observational data that greater protein delivery can cause harm. A major cohort study from the UK published in JAMA reported a greater loss of quadriceps muscle – diagnosed on ultrasound – in those that received more dietary protein[15]. In addition, in a pre-planned observational study of 1440 critically ill children within a large randomised clinical trial of nutritional therapy, increased protein was associated with worse outcomes, e.g., increased protein increased the risk of infections and reduced the likelihood of being liberated from mechanical ventilation and discharged alive from ICU[16].

Clinicians/researchers who advocate for less enteral protein administration propose that dietary protein impairs the normal 'autophagy' process[17]. Autophagy is the pathway to clear damaged organelles and proteins from muscle and organs[18]. This pathway of autophagy is crucial for recovery from critical illness. This concept was supported by a study of 36 critically ill patients that demonstrated a strong linear relationship between increased protein delivery and impaired cellular markers of autophagy[19]. Animal models of critical illness also report that protein markedly attenuates autophagy[20].

Current clinical practice

There is a clear dissociation between international guidelines and current clinical practice. Survey, point- and period- prevalence data conducted by our group indicate that Australian ICU patients, similar to patients world-wide, do not receive the amount of protein that is recommended in the international clinical practice guidelines. Rather patients receive between 0.6 to 1.0 g/kg/day of protein[21]. It should be noted that this range is similar to the quantity recommended in health (~0.8 g/kg/day).

Systematic review and meta-analysis using data from randomised clinical trials

We recently published a systematic review comparing randomised trials that had one group of adult ICU patients who received enteral protein to amounts recommended in international clinical guidelines (≥ 1.2 g/kg/day) and another group who received enteral protein to amounts representative of usual care (< 1.0 g/kg/day). There were only five small, single-centre, randomised clinical trials that met our selection criteria with a total of 431 patients enrolled. All trials had a substantial risk of bias. The meta-analysis suggested that augmenting protein reduced mortality[22] (**Figure 1**).

Feasibility trial comparing augmented enteral protein to usual care

We subsequently completed an investigator-initiated, binational, multicentre, parallel group, blinded, feasibility trial of augmented protein to usual care in critically ill adults receiving invasive organ support (HREC Reference number: HREC/18/CALHN/658; Australian New Zealand Clinical Trials Registry: ACTRN 12618001829202)[23]. Nutricia Research B.V. provided two liquid formulae that were indistinguishable in colour, viscosity and packaging. One formula contained augmented protein (100 g/1000 mL, administered to the intervention group) and the other the usual amount of protein (63 g/1000 mL, administered to the control group). We were able to successfully manage logistics, with a subsequent analysis of phenylalanine content of trial formulae confirming the correct delivery of the allocated solutions in all samples. We enrolled 116 adult participants across 6 ICUs within 3 months. The group with the augmented formula received protein quantity within international guidelines whereas the other received protein quantity representative of usual care (1.52 (0.52) vs. 0.99 (0.27) g/kg ideal body weight (IBW)/day; group difference 0.53 (95% CI 0.38 to 0.69)). Trial formula was delivered for a median of 8.5 days with similar calorie delivery throughout. The difference in mortality favoured augmented protein delivery (difference between groups using generalised estimating equation -1.3% (95% CI -17.7 to 15.0%))[23].

A large high-quality trial is now needed

Based on the observational data, meta-analysis, our single-centre mechanistic study, and our multi-centre feasibility trial, we submit that a large trial is warranted to determine whether augmenting dietary protein to levels recommended in international guidelines improves outcomes that are important to patients. Enteral nutrition is an extremely inexpensive therapy, so any reduction in mortality and/or morbidity will represent a highly cost-effective intervention.

5. RESEARCH HYPOTHESES

Our primary hypothesis of the TARGET Protein trial is that augmenting dietary protein to critically ill adult patients when compared to usual care will increase the number of days free of the index hospital and alive at day 90.

Our secondary hypotheses are that augmented dietary protein to critically ill adult patients when compared to usual care:

1. Reduces all-cause 90-day mortality
2. Reduces duration of mechanical ventilation
3. Reduces duration of admission to ICU and hospital
4. Reduces the incidence of tracheostomy
5. Does not increase the incidence of new renal replacement therapy
6. Increases days alive and out of the index hospital in those who survive at least 90 days

6. OUTCOMES

Primary outcome

- Number of days free of the index hospital and alive at day 90.

  This will be calculated as 90 days and subtracting all days admitted to the index hospital after commencement of trial enteral formula, plus any days readmitted to the index hospital within 90 days. As patients may be readmitted for part of a day (e.g. for dialysis) a readmission will only be counted when the patient remains in hospital at 2400 hours. Patients will be considered alive if they are alive at discharge from the index hospital, and there is no evidence of death before day 90. In Australia, death after hospital discharge will be ascertained using data linkage with the Australian National Death Index[24, 25]. At the New Zealand site, the Adult Patient Database is linked to the New Zealand national death registry to record death after hospital discharge. Patients who die during the 90-day period, will be assigned a value of 0.

Secondary outcomes

- Days free of the index hospital at day-90 in survivors

  This will be calculated only for those that survived to day 90. The outcome will be the same as the primary outcome and calculated as 90 days and subtracting all days admitted to the index hospital after commencement of trial enteral formula, plus any days readmitted to the index hospital within 90 days. As patients may be readmitted for part of a day (e.g., for dialysis) a readmission will only be counted when the patient remains in hospital at 2400 hours
- Alive at day-90

  Patients will be considered alive if they are alive at discharge from the index hospital, and there is no evidence of death before day 90. Health service records will be used to provide evidence of patient status, including if the patient was discharged alive from hospital but known to have a health event prior to day 90. Health events include rehabilitation hospital admission, outpatient appointment attended, pathology test completed, other medical investigation, day admission eg dialysis, chemotherapy.
- Duration of invasive ventilation (hours)
- Duration of admission to ICU (days)
- Duration of admission to hospital (days)
- Incidence of tracheostomy insertion
- Incidence of renal replacement therapy
- Discharge destination
- Health economic analysis

7. RESEARCH PLAN

Design

A cluster randomised, cross-sectional, double cross-over, registry-embedded, pragmatic clinical trial

Study Participants

We will conduct the study at 8 high volume ICUs across Australia and New Zealand

Inclusion criteria
1. ≥16 years of age
2. Receiving/about to commence EN during their index admission to ICU or receiving/about to

	commence EN for the first time in ICU during subsequent admissions
Exclusion criteria	
1.	At the commencement of EN, the treating clinician considers the trial enteral formula to be contraindicated
2.	The patient has received greater than 12 hours of non-trial enteral formula
3.	The patient has previously participated in the TARGET Protein trial

Randomisation

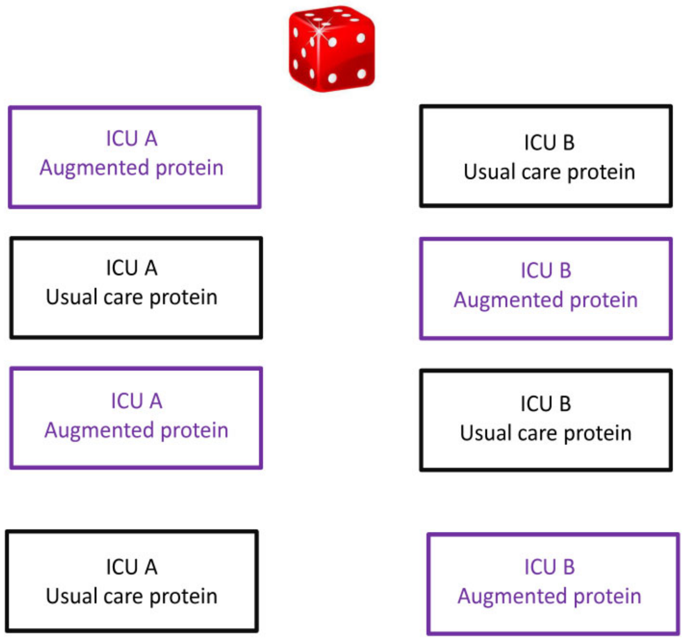
All patients in a given ICU (cluster) who meet eligibility criteria will receive the same formula across a 3-month period. The formula assigned to the ICU (cluster) for that period is the variable randomised. This allows the intervention to be delivered as if it were usual care, which increases the efficiency of data collection and reduces the burden on participating sites. After a 3-month period, the ICU will then administer the alternative formula for all patient admissions over the next 3 months. Participants will continue to receive the treatment that they were originally assigned if they remain in the ICU during a crossover period. The process is then repeated so each ICU (cluster) crosses over twice (**Figure 2**).

All patients receiving enteral nutrition at each site will receive the formula to which the site is randomised during the current cluster period, unless a speciality formula is required (i.e. elemental formula).

A sensitivity analysis of the primary result will be completed, and patients will be excluded from the analysis if they have been transferred from another ICU or are known to have received non-study enteral nutrition for greater than 12 hours.

Similar to a parallel-group trial, the inclusion of the crossover element increases the power of the study and reduces the required number of participants when compared to a parallel approach[26]. In standard cluster trials, the sample size estimation requires modification for differences that occur between clusters. However, in TARGET Protein, the multiple cross-over periods for each cluster reduces the differences within cluster periods (which are termed the intra-class coefficient), thereby increasing power of the trial[26]. Four cluster-periods of 3 months equates to each ICU participating for 12 months (Figure 2).

Our group also recently published a cluster randomised, single cross-over, clinical trial that was conducted across 5 countries to compare in-hospital mortality rates using proton pump inhibitors to a histamine receptor blockade for stress ulcer prophylaxis[27]. In this drug trial, 26,982 patients were randomised in 12 months of active recruitment per site. As this trial was open label, no washout period was included.



Study Treatments

The study enteral nutrition (EN) (intervention and control) will be provided to sites by Nutricia Research B.V.

***Intervention (augmented dietary protein group)*:** **'**Nutrison Protein Intense' 1.26 kcal/ml and 100g protein per 1000 ml (**Appendix 1**).

Control (usual dietary protein group): 'Nutrison Protein Plus' 1.25 kcal/ml and 63g protein per 1000ml (**Appendix 2**).

The protein sources are similar between formulae, each derived from whey, casein, pea, and soy protein. The calorie content (1.26 vs. 1.25 kcal/ml), fat content (49g/1000 ml), and osmolality (340 mOsmol/kgH_2O) in both formulae are also similar and both formulae are fibre-free.

Study procedures

Following the treating clinician's decision to commence EN, the participant will receive the study EN formula to which the ICU is currently randomised.

The goal rate for study EN will be as per usual site processes with the maximum goal rate set by the site's treating clinician (intensivist or dietitian) as per usual clinical care. To ensure patient safety and prevent excess delivery, sites will receive further education regarding the use of prescribing nutrition to ideal body weight.

All aspects of nutrition management, other than the choice of formula, will be according to individual unit practice. The rate at which study EN is commenced and incremented, and strategies to increase nutrient delivery (e.g. pro-kinetic drugs, post-pyloric tubes) in both treatment groups, will be at the discretion of the treating team usual unit nutrition protocol.

Study EN will be administered when clinically indicated until either the patient reaches day 90, is discharged from ICU, or dies. Patients discharged and readmitted to the ICU within 90 days of study enrolment still requiring EN will be recommenced on study EN as per the previous treatment allocation.

Discontinuation of study EN and/or withdrawal:

The study EN will cease if one of the following occur:

1. Cessation of EN by the treating clinicians before day 90. If EN is recommenced before day 90, study EN should be resumed as per the previous treatment allocation unless treating clinician feels that it is in the patient's best interest to cease the study intervention.
2. Patient is discharged from the hospital before day 90. For patients discharged from ICU and readmitted within 90 days within the index admission they will receive the study EN originally allocated.
3. Treating clinician feels that it is in the patient's best interest to cease the study EN.
4. Patient dies prior to day 90.

Co-enrolment

As this study represents standard care, co-enrolment in other trials is encouraged and will be considered by the TARGET Protein Management Committee when requested. Studies involving another nutrition intervention in ICU for >24hours will be reviewed by the TARGET Protein Management Committee on an individual basis. The TARGET Protein Management Committee approved the co-enrolment of patients into the INTENT nutrition trial (ClinicalTrials.gov Identifier: NCT03292237) [28]. INTENT is an open label, randomised controlled trial aiming to optimise energy delivery using supplemental parenteral nutrition in ICU and an intensive nutrition intervention in the post ICU period. The primary outcome of the INTENT Trial is daily energy delivered rom nutrition therapy. Two of the eight TARGET Protein sites are also participating in INTENT. Co-enrolment was approved on the basis that the number of patients co-enrolled was likely to be few (< 5) and unlikely to impact the validity of either trial.

8. DATA MANAGEMENT

The project data management conforms to the NHMRC guidelines published in 2019 'Management of Data and Information in Research: A guide supporting the Australian Code for the Responsible Conduct of Research'.

The local site principal investigators will be responsible for data management at participating sites. The TARGET Protein chief investigator and the coordinating centre (Royal Adelaide Hospital) will be responsible for the management of pooled non-identifiable data.

8.1 Data category:

Re-identifiable: A master enrolment log of participants will be completed for the study at each participating site. This will record identifying details of each participant and will be used to assign each participant a study number. The enrolment logs will be retained securely and confidentially at each participating site. The investigators propose that the data obtained remain re-identifiable to enable the data to be interrogated/corrected in the case of unexpected aberrant data e.g. database and data entry errors. Re-identifiable data may be used for data verification and query resolution and only referred to, if required, by local site investigators.

8.2 Data storage:

The local site principal investigators will be responsible for data storage at the participating sites. Data will be collated, anonymised and entered into a secure password protected web database electronic Case Report Form at participating sites. Sites will store enrolment logs and non-identifiable (anonymised) participant data according to local site data management policies.

Pooled (all sites) anonymised data will be stored at the coordinating centre as a secure password protected electronic file.

8.3 Data access:

The Chief Investigator is responsible for the data overall. Each participating centre retains ownership of their local site data managed by the principal investigator. Only those persons directly involved in data collection and analysis will have access to site study data. The registry data from each site comes from the Australian New Zealand Intensive Care Society Adult Patient Database (ANZICS APD) that each site already collects, records and owns. Each site will arrange secure data transfer via an encrypted file containing a de-identified subset of data from the ANZICS APD to the coordinating centre.

The pooled anonymised data will be held at the coordinating centre and remain the property of the Target Protein Management Committee and only study investigators will have access to stored non-identifiable pooled electronic data. As required by many high impact journals the investigators will have to include a data sharing agreement. The investigators propose to use the following statement in the final manuscript. "Non-identifiable individual participant data that underlie the results reported in this trial will be made available after 3 years following publication and ending 7 years after publication. Availability will only be made to researchers who provide a written proposal for data evaluation that is judged to be methodologically sound by a committee approved by the TARGET PROTEIN Investigators. Proposals should be directed to targetprotein@adelaide.edu.au. If the proposal is approved, access data requestors will be required to sign a data access agreement prior to accessing data.

8.4 Data retention and destruction:

Both electronic and hard copy study data will be retained for a period of fifteen years following completion of the study. Following that time, information stored will be disposed of in a confidential manner as per site policy.

In addition to the processes described above, the investigators acknowledge that data may otherwise be discoverable through processes of law or for assessing compliance with research procedures.

8.5 Data collection

The Australian New Zealand Intensive Care Society Adult Patient Database

The Australian New Zealand Intensive Care Society Adult Patient Database (ANZICS APD) was set up in 1992 and currently holds > 2 million patient episodes[29]. The APD includes data from every patient admitted to an Australian ICU. Demographic and physiological data are collected as part of routine clinical practice on admission to ICU and submitted to the ANZICS APD, supporting benchmarking and quality assurance processes. These data allows a comprehensive table of baseline characteristics to be generated in the eventual manuscript[29].

Data extracted from the ANZICS APD will include (**Appendix 3** APD minimum data set):

1. Baseline demographics (age, sex, weight, height)
2. Baseline variables at ICU admission:
    a. Acute Physiology And Chronic Health Evaluation (APACHE) II score
    b. ICU admission diagnosis from APACHE III score, Elective Emergency or non-surgical category
    c. Ethnicity
    d. Clinical frailty score within 2 months preceding their first admission to ICU within the current hospital stay
    e. Biochemistry data (highest and lowest within 24h of ICU admission): urea, creatinine, albumin, potassium, blood glucose
3. Ventilator type on ICU admission (day 1) (non-ventilated, ventilated, ventilated excluding cardiac surgery)
4. Post intervention outcome data including:
    a. Mechanical ventilation hours in ICU;
    b. Tracheostomy insertion in ICU;
    c. Need for renal replacement therapy in ICU;
    d. ICU and hospital source of admission;
    e. Duration of ICU and hospital admission;
    f. ICU and hospital mortality;
    g. ICU readmissions within the same hospital admission;
    h. ICU and hospital readmissions to the index hospital;
    i. Discharge destination (home, another acute hospital, rehabilitation, nursing home/chronic/palliative care, ICU in another hospital, other destination).

If a site does not collect the above as ANZICS APD minimum data set, the data will be collected as additional data.

Additional data collected as part of the study will include:

1. Daily volume of EN delivered, collected on calendar days 1-5, day 10, day 20, day 30 and day 90; (Study day 1 commences on the day TARGET Protein trial enteral formula is commenced and concludes at the end of the calendar day (2400h), i.e., will represent a partial day).
2. Biochemistry data: Blood urea, serum albumin, serum phosphate, and blood glucose closest to 0800h on study days 1-5, and day 10.
3. Adverse events collected whilst participants are receiving study EN up until 48 hours post cessation of study EN.
4. ICU and hospital readmissions to the index ICU/hospital

5. Last known patient status at Day 90 as per medical records - At any time up to study day 90 as per medical records to ascertain whether the patient is alive at day 90 without direct contact.

Study day 1 commences on ICU admission and concludes at the end of the calendar day (2400h).

For the primary outcome 'Days free of the index hospital and alive at day-90' we will rely on registry data with data linkage to the National Death Index. This approach is consistent with our application for waiver-of consent as we will not contact the patient after discharge.

To check site compliance with provision of the correct enteral nutrition to which the ICU has been randomised during a given cluster period, we will conduct a random audit. The audit will require site research staff visualising which feed is hung closest to 8am on randomly allocated days. These days will be during a cluster period and close to cluster change over. The need to visualise the feed hung at the bedside is required as our impression is that medical record documented formula names may not reflect which feed is actually hung at the bedside.

8.6 Data entry

A secure password protected internet-based case report form will be used to record all patient data - Research Electronic Data Capture (REDCap) software. REDCap is a secure Good Clinical Practice-compliant web-based server. Study participants will be followed up until either death or until 90 days after enrolment, censored at hospital discharge.

8.7 Data management and data quality

Participant eligibility criteria were validated in our feasibility trial where a potential survival benefit with no signal for harm was identified[23]. Monitoring of study sites and case report forms will be conducted to ensure consistency of data collection and compliance with study protocol. Due to the rapid recruitment (12 months) an interim analysis for early efficacy will not be conducted. Regardless, an independent Data and Safety Monitoring Committee (DSMC) will be formed with experts in intensive care medicine, intensive care nutrition, statistics and consumer safety to review protocol compliance, serious adverse events and baseline mortality on a 3-monthly basis. The DSMC will advise whether the study should be amended, stopped or continue unchanged.

8.8 Safety

Adverse events

It is recognised that the patient population in the ICU will experience a number of aberrations in laboratory values, signs and symptoms due to the severity of the underlying disease and the impact of standard treatments in the ICU. These will not necessarily constitute adverse events or serious adverse events unless they are considered to be related to study treatment or in the principal investigator's clinical judgement are not recognised events consistent with the patient's underlying critical illness and/or chronic diseases and expected clinical course.

Accordingly, the reporting of adverse events will be restricted to events that are considered to be related to study treatment (possibly, probably or definitely).

Adverse events (AEs) are defined as any untoward medical occurrence in a patient or clinical investigation subject administered an investigational intervention and does not necessarily have to have a causal relationship with this intervention (adapted from the Note for Guidance on Clinical Safety Data Management: Definitions and Standards for Expedited Reporting (CPMP/ICH/377/95 July 2000). In all cases, the condition or disease underlying the symptom, sign or laboratory value should be reported e.g. renal failure rather than hyperkalaemia, and agitation rather than self-extubation.

AEs will be collected from commencement of study EN up until 48 hours post cessation of study EN only. The patient should be followed until the event is resolved or explained. Frequency of follow-up is left to the discretion of the investigator.

Serious adverse events

The baseline mortality of intensive care patients is high due to the illness that has necessitated their ICU admission. Intensive care patients will frequently develop life-threatening organ failure(s) unrelated to study interventions and despite optimal management. Therefore, consistent with ICU international best-practice and, as outlined above, deaths that are part of the natural history of the primary disease process or expected complications of critical illness will not be reported as serious adverse events in this study[30]. In particular, events already defined and reported as study outcomes (e.g. mortality) will not be reported separately as adverse or serious adverse events unless they are considered to be causally related to the study intervention or are otherwise of concern in the investigator's judgement. Serious adverse events will be reported up to day 90 post-study inclusion for the list of events below that are considered to be related to enteral nutrition. These are:

- Mesenteric ischaemia
- Abdominal compartment syndrome

Reporting adverse events

Adverse events will be recorded in the eCRF. All serious adverse events must be reported to the coordinating centre via the study database within 72 hours of the investigators becoming aware of the event.

The site principal investigator will be responsible for notifying the Institutional / Ethics Committee of the occurrence of any serious adverse event likely to affect the safety of patients or the continued conduct of the clinical trial, in particular any change in safety. Serious adverse events will also be reported to the study EN manufacturer, Nutricia Research B.V.

9. ANALYSIS AND REPORTING OF RESULTS

9.1 Statistical considerations and sample size

Sample size has been calculated using The Shiny Cluster Randomised Trial Calculator (https://clusterrcts.shinyapps.io/rshinyapp/) and data from TARGET Calories[2]. In TARGET Calories the standard deviation for the primary outcome we will use in TARGET Protein (days alive and out of hospital censored at day 90) was 11.3 days. Assuming the coefficient of variation for cluster size is 0.5 – based on 6 ICUs (i.e. 6 clusters) each enrolling a mean of 160 patients/3 month period (95% confidence interval, 80 to 240 patients per cluster per period) - and the default settings for cluster trials of a intracluster correlation coefficients of 0.02 (with a lower extreme of 0.001 and upper extreme of 0.1) 60 to 65 patients per period per cluster would provide 80% power, and ~80 patients per cluster per period would provide 90% power to find a difference of 1 day in days alive and out of hospital to day 90. Assuming a parametric distribution a sample size of 2560 would provide 90% power. However, the primary outcome (days alive and out of hospital) is unlikely to be normally distributed and parametric calculations are optimistic. For this reason, we have inflated the sample by 15% to allow for the likely skewed distribution of data, such that ~3,000 patients should provide 90% power. We are assured of enrolling at least 3,000 participants in a 12-month period as we will have 8 high volume ICUs as clusters.

9.2 Analysis plan

. For the primary outcome (days free of the index hospital and alive at day 90) and the first secondary outcome, will use individual patient-level data and fit a quantile mixed effects model to compare the median response between treatment groups. The mixed effects model will include treatment group, period, and the stratification variable used in randomisation, delayed start ("Group 1 – Commencing May 2022" and "Group 2- Commencing August 2022"), as fixed effects. Cluster (ICU) will be included as a random effect and assumed to be normally distributed with mean zero and variance components σ_C^2 (5). The effect of treatment comparisons will be presented as a difference in medians (95% confidence interval (CI)); the 95% CIs will be calculated using the block-bootstrap method. For the binary secondary outcomes (i.e., alive at day 90, incidence of tracheostomy insertion and incidence of renal replacement therapy), we will fit generalised estimating equations (GEE) with a logarithmic link function, an exchangeable working correlation matrix and robust standard errors using the ICU as the clustering unit. Finally, time to event secondary outcomes (i.e. duration of invasive ventilation, admission to ICU and admission to hospital), will be analysed by fitting Cox regression models with covariate adjustment for period and delayed start, with stratification by ICU, and robust standard errors clustered at ICU level (for any residual within-ICU correlation) to estimate cause-specific hazard ratios and confidence intervals, with patients dying prior to discharge (or extubation) censored at their time of death. Sensitivity analyses will be conducted to investigate the impact of missing data and competing risk of death. Sub-group analyses will be presented for the following subgroups: invasive mechanical ventilation, age 70 years and older, body mass index of 35 kg/m^2 or above and acute kidney injury (AKI) prior to receiving trial entera formula Following publication of the EFFORT trial [31], acute kidney injury on admission to ICU was added as a sub-group. Acute kidney injury is defined as receiving new renal replacement therapy between hospital admission and commencement of trial enteral formula The protocol and statistical analysis plan with analysis code will be pre-published[32, 33].

10. ETHICAL CONSIDERATIONS

10.1 Ethical conduct

TARGET Protein will be performed in accordance with all relevant ethical and regulatory approvals and guidelines laid down by the National Health and Medical Research Council (NHMRC) of Australia and the Ottawa statement on the Ethical Design and Conduct of Cluster Randomised Trials[34], the Declaration of Helsinki, its subsequent amendments, and the ICH GCP guidelines on the ethical conduct of research.

Ethics applications will be submitted to the relevant ethics committees of the sites participating; consisting of dual applications in both New Zealand and Australia in accordance with the relevant jurisdictional laws. All Australian TARGET Protein sites are participants in National Mutual Acceptance system for single scientific and ethical review. Each principal investigator will be responsible for applying for site authorisation through the existing research governance systems in the jurisdiction.

10.2 Ethical implications of a cross over cluster randomised control design

The rationale for the use of a cluster randomised crossover (CRXO) trial is that this is the most appropriate method to answer the research question. The effect size of the intervention, augmenting dietary protein, is relatively small and a large number of participants would be required using a traditional two arm randomised control trial (RCT) making recruitment prohibitive, even across multiple intensive care units. Employing a CRXO design enables recruitment at the institutional level with reduced burden on participating sites in regard to study logistics and data collection and allowing for a larger number of participants recruited at a feasible cost from a funding perspective. Perhaps most importantly, by reducing logistics it reduces risk of incorrect formula being administered – which has implications for individual patient safety and trial integrity. Finally, as feeding protocols vary between ICUs the cross-over component of a cluster randomised cross sectional trial ensures variability and intraclass coefficient is reduced. Accordingly, our design has similar power as a parallel group trial at markedly reduced cost with greater safety.

As the intervention will be given to all eligible patients in participating ICUs as part of 'usual practice' for that ICU, large numbers of patients will participate and the investigators propose that the most suitable and feasible method of enrolment in the study will be without individual patient consent. Additionally, it is not possible to identify and consent potential cluster participants prior to unit randomisation due to the very nature of ICU patients' emergency admission status.

10.3 Ethical issues specific to TARGET Protein participants

The NHMRC National Statement on Ethical Conduct in Human Research acknowledges, in chapter 4.4, that research involving people highly dependent on medical care who may be unable to give consent, such as the participants in the proposed study, are necessary to assess and improve the efficacy and safety of interventions used in their treatment. The TARGET Protein protocol is consistent with the values, principles and themes set out in the guidelines. The results of TARGET Protein will increase the understanding of the role dietary protein plays in ICU patients' recovery from critical illness.

The majority patients eligible to be enrolled in this study will lack the capacity to consent for themselves at the time of enrolment due to their underlying medical condition requiring unplanned/emergency life support measures and an admission to an ICU. The investigators acknowledge that explicit prior consent for research would be ideal in this vulnerable patient group;

however, the unplanned/emergency life support measures required mean it will not be possible or appropriate to seek prior informed consent from participants or next of kin to cluster randomisation or study enrolment. The majority of study participants will be unconscious at the time of initiation of enteral nutrition, and hence their wishes cannot be determined. A small number of participants admitted to ICU who are not mechanically ventilated or sedated but require enteral nutrition will also be included. However, this patient population will be critically unwell and with an altered conscious state and therefore unlikely to be able to provide informed consent for themselves.

The NHMRC statement section 4.4.13 (a-f) identifies issues pertaining to participation in a research project without prior consent. The Investigators propose that this study complies with these issues:

4.4.13 When neither the potential participant nor another on his or her behalf can consider the proposal and give consent, an HREC may, having taken account of relevant jurisdictional laws, approve a research project without prior consent if:

(a) there is no reason to believe that, were the participant or the participant's representative to be informed of the proposal, he or she would be unwilling to consent; The Investigators have no reason to believe that critically ill patients in ICU would not be willing to consent to this low risk study if asked prior.

(b) the risks of harm to individuals, families or groups linked to the participant, or to their financial or social interests, are minimised; TARGET Protein is a low risk study evaluating administration of enteral protein quantities that fall within that recommended by leading international enteral nutrition societies[13, 14] and comparing this to current usual care. It is currently unknown which of these protein quantity/prescription is optimal for critically ill patients; however, we have preliminary evidence (from a single-centre trial, systematic review and meta-analysis and our multi-centre feasibility trial) that augmented dietary protein may improve outcomes. Accordingly, neither dietary approaches in this research expose participants to protein delivery outside of usual care or that recommended by international guidelines. The study enteral nutrition are commercially available and are used as part of usual clinical care in Australia and New Zealand. There are no additional requirements for the study, all other aspects of clinical care will remain the same. The study will use existing registry data on patient demographics, clinical care and outcomes, which are documented for quality assurance purposes across Australia and New Zealand. In addition, a small number of data will be extracted from the participants' medical records. These data are necessary in order to ascertain the protein quantities received by participants and are documented as a part of routine clinical care at all sites (i.e. volume of enteral nutrition delivered). All data collected for the study will be anonymised to protect patient confidentiality and privacy.

(c) the project is not controversial and does not involve significant moral or cultural sensitivities in the community; While augmenting dietary protein is not a controversial intervention we have established that the mean amount of protein delivered is much less than guideline recommendation. This probably reflects clinician indifference due to a lack of data from well-conducted randomised clinical trials rather any moral or cultural sensitivity. In addition to the study intervention, participants will continue to receive usual ICU care as per the treating clinician regardless of study involvement. The study does not involve significant moral or cultural sensitivities in the community.

and, where the research is interventional, only if in addition:

(d) the research supports a reasonable possibility of benefit over standard care;

An ICU stay is associated with substantial mortality, with 30% of patients dying within 3 months of admission. In addition, significant muscle wasting occurs in up to 80% of critically ill patients, resulting in 'ICU-acquired weakness' which adversely affects outcomes from critical illness. The provision of additional dietary protein has the potential to improve survival, attenuate muscle atrophy, and hence improve outcomes following critical illness. TARGET Protein will lead to an increased understanding of nutrition support practices in critically ill patients to inform international nutrition guidelines and hence lead to an improvement in care for critically ill patients.

e) any risk or burden of the intervention to the participant is justified by its potential benefits to him or her; As discussed above, TARGET Protein is a low risk interventional study evaluating protein prescriptions that fall within that recommended by leading international nutrition societies[13, 14] and comparing this to current usual care and data documented for the study will have been already collected for clinical care and quality assurance purposes across Australia and New Zealand.

(f) inclusion in the research project is not contrary to the interests of the participant. This low risk study eligibility criteria specifically exclude patients in which enrolment is not considered to be in the participant's best interests.

10.4 Waiver of consent

Participation in TARGET Protein is no more than low risk to participants. We have carefully considered the provisions of chapter 2.3 of the NHMRC National Statement, which addresses the issues pertaining to qualifying or waiving conditions for consent. The TARGET Protein investigators believe TARGET Protein complies with these and that neither explicit consent nor an opt out approach are feasible or appropriate. The Investigators are requesting HREC approval for a waiver of consent to enrol participants in the TARGET Protein study according to guidelines set out in section 2.3.10 and 2.3.12:

2.3.10 (a) involvement in the research carries no more than low risk to participants; This study is comparing two protein prescriptions using enteral formulae with differing protein content that are frequently used in ICU patients in Australia (i.e. the protein prescriptions being compared are neither new nor experimental). The TARGET Protein study is low risk to participants, as both dietary approaches are acceptable as current patient management as recommended in international critical care nutrition guidelines. Neither dietary approach in this study expose participants to protein doses outside of international guideline recommendations. Moreover, we have previously shown in our multi-centre feasibility trial that we can safely deliver protein doses within international recommendations using the interventional formulae and protein doses consistent with current Australian practice using the usual care formulae (intervention: 1.52 (0.52) vs. usual care: 0.99 (0.27) g of protein/kg ideal body weight (IBW)/day; group difference 0.53 (95% CI 0.38 to 0.69)). Furthermore, the difference in mortality in this feasibility trial favoured augmented protein delivery (difference between groups using generalised estimating equation -1.3% (95% CI -17.7 to 15.0%). We submit that allowing withdrawal of enrolled participants (such as with a delayed or an 'opt-out' approach) may introduce bias into mortality estimates, which could affect the accuracy of mortality estimates, potentially harming future participants.

2.3.10 (b) the benefits from the research justify any risks of harm associated with not seeking consent; The Investigators have exercised beneficence designing the TARGET Protein protocol with maximum benefit and harm minimisation to participants at the forefront. For example, the interventional therapy falls within current guidelines, the data management plan protects participant privacy and identity, safety event reporting guidelines will be followed, and a study monitoring plan will be in place.

The potential benefit of participation for the individual is that preliminary evidence supports the concept that the intervention dietary approach (increasing protein) may improve outcomes when compared to current usual care. TARGET Protein will lead to an increased understanding and an improvement in care of critically ill patients and inform international guidelines on enteral nutrition in this patient group. This may lead to improved outcomes of critically ill patients and reduced hospital/health care costs.

2.3.10 (c) **it is impracticable to obtain consent (for example, due to the quantity, age or accessibility of records)** Prior explicit consent for study participation cannot be obtained because eligible participants will be receiving unplanned life support in the ICU. The majority of patients eligible for this study will be unable or have an impaired ability to consent upon admission to the ICU (due to medical condition or life sustaining treatment, or consciousness), and a next of kin may not be available. The informed consent process with next of kin of critically ill patients is lengthy and can be burdensome for already distressed relatives. It is proposed that approaching next of kin for consent for a low-risk study soon after patient admission to ICU is an unnecessary additional stress that can impede comprehension of their next of kin already complex condition and care. In Australia and New Zealand usual ICU practice is that enteral nutrition is considered an important treatment to be commenced as soon as practicable after tracheal intubation or where oral intake is not possible, when there are significant time constraints. Based on international guidelines, it is believed that delaying the initiation of feeding, particularly in poorly nourished patients, will impact negatively on patients' outcomes[13].

TARGET Protein is a large low risk study. Given the low risk and that we are collecting registry data we submit that it not feasible to obtain consent to retain data from such large numbers of patients and/or their relatives (estimated n= >600 participants in each centre over 12 months). The sites research team do not have the capacity or resources to coordinate and carry out approximately 20-30 consents discussions per week solely for the purpose of retaining already collected data.

2.3.10 (d) there is no known or likely reason for thinking that participants would not have consented if they had been asked; There is no known or likely reason for thinking that participants would not have consented if they had been asked. Due to the nature of the intervention (dietary protein, as described above), with minimal-low risk and the collection of minimal data, which is already documented for clinical use and entered into electronic medical records and a centralised ICU database, we have no reason to believe that ICU patients would not have consented to this research.

2.3.10 (e) there is sufficient protection of their privacy (f) there is an adequate plan to protect the confidentiality of data; At all times ICU patients' privacy and confidentiality will be respected and upheld. The study protocol data management plan protects the identity of participants (see section 8 and 10.5). The local site principal investigators retain overall responsibility to ensure site compliance with the protocol data management plan. Research staff involved with TARGET Protein will be employed by site institutions and as such, comply with the relevant jurisdictional laws and the organisational procedures and policies. Additionally, participants will not be identifiable in any form within any presentations or publications arising from the research.

2.3.10 (g&h) not applicable

2.3.10 (i) the waiver is not prohibited by State, federal, or international law. All Australian TARGET Protein sites are participants in the National Mutual Acceptance system for single scientific and ethical review. A waiver of consent is not prohibited by law at these sites. The site principal investigators will be responsible for local Governance approval and study conduct in accordance with local jurisdictional laws and institutional policies. A separate HREC application and approval process will be conducted in New Zealand specific to local policies, requirements and laws.

2.3.12 Given the importance of maintaining public confidence in the research process, it is the responsibility of each institution to make publicly accessible (for example in annual reports) summary descriptions of all its research projects for which consent has been waived under paragraph 2.3.10; Registration of the study will be made via the Australian New Zealand Clinical Trials Registry, which is a publicly available online registry of clinical trials being undertaken in Australia and New Zealand. On this registry summary description of the study details is provided.

South Australian sites only - Consent waiver for screening activities:

The investigators are proposing that a waiver of consent for screening of ICU patients via access to the patients' medical records for the TARGET Protein study is the most applicable process for screening and eligibility assessment. The treating clinicians, principal investigators and the ICU research lead have delegated the intricacies of screening for the TARGET Protein study to the appropriately trained ICU research staff employed by CALHN.

It is not always feasible or appropriate to discuss a low risk study screening activities with potential participants or their next of kin to obtain consent to access medical records. The screening process is a low and negligible risk process part of research. The NHMRC National Statement section 2.3.10 (a-f & i) addresses the issues pertaining to a waiver of consent process. The investigators believe that this study protocol complies with these issues and are requesting approval for South Australian sites a waived consent process for screening and eligibility assessment.

2.3.10a Involvement in the research carries no more than low risk to participants.
Screening involves accessing electronic and paper records to determine eligibility for inclusion in approved research studies and with appropriate measures in place to preserve privacy and confidentiality and is a negligible risk procedure. As such, care of participants medical condition is unaffected by this.

2.3.10b The benefits from the research justify any risks of harm associated with not seeking consent:
There is no anticipated risk of harm from accessing medical records to complete screening / eligibility assessments. Although there is no direct benefit to the participant the project has the potential to

benefit the health care of critically ill patients by providing additional information regarding an existing or new treatment.

2.3.10c It is impracticable to obtain consent: Due to the significant number of admissions to the ICUs it is not feasible for the clinical team to obtain consent for screening/ eligibility assessment from all critically ill patients or their relatives for clinical research studies. The ICU clinicians believe it would be both inappropriate and impracticable to obtain consent for the purposes of screening. Further, many patients have impaired ability to consent upon admission to the ICU (due to medical condition, or consciousness), and a decision maker may not be available.

2.3.10d There is no known or likely reason for thinking that participants would not have consented if they had been asked: The treating clinicians and investigators have no reason to believe that participants would not have consented to data access for screening activities if asked. Accessing medical records to determine eligibility will ensure that patients are not unnecessarily enrolled into a study that they are ineligible for.

2.3.10 e & f Privacy and Confidentiality: At all times ICU Patients' privacy and confidentiality will be respected an upheld. ICU Research staff are employed by CALHN and as such comply with the organisational requirements; SA Health job and person specifications acknowledge:

"SA Health employees will not access or attempt to access official information, including confidential patient information other than in connection with the performance by them of their duties and/or as authorised. SA Health employees will not misuse information gained in their official capacity. SA Health employees will maintain the integrity and security of official or confidential information for which they are responsible. Employees will also ensure that the privacy of individuals is maintained and will only release or disclose information in accordance with relevant legislation, industrial instruments, policy, or lawful and reasonable direction."

The Investigators believe that the study data management process protects the identity of screened patients and participants. Screening information is important to later analyse generalisability of findings and describe the population; screening logs will not contain any patient identifiers and will be housed along with the study document files for the required duration for the study.

2.3.10 (g&h) not applicable

2.3.10i The waiver is not prohibited by State, federal, or international law: A waiver of consent is not prohibited in South Australia by State, federal, or international law.

Victorian sites only

The TARGET Protein protocol is consistent with state laws at the proposed site.

As per the requirements of the national mutual recognition scheme we have completed the Victorian Specific Module to ensure that the trial complies with Victoria law as set out in the *Medical Treatment Planning and Decision Act 2016.*

As per the *Medical Treatment Planning and Decision Act 2016* a medical research procedure is defined as (Part I, page 6-7)

"Medical research procedure means—
(a) a procedure carried out for the purposes of medical research, including, as part of a clinical trial—
(i) the administration of pharmaceuticals; or
(ii) the use of equipment or a device; or
(b) a prescribed medical research procedure—

but does not include any of the following—
(c) any non-intrusive examination including—
(i) a visual examination of the mouth, throat, nasal cavity, eyes or ears; or
(ii) the measuring of a person's height, weight or vision;
(d) observing a person's activities;
(e) undertaking a survey;
(f) collecting or using information, including either of the following—
(i) personal information within the meaning of the Privacy and Data Protection Act 2014;
(ii) health information;
(g) any other procedure prescribed not to be a medical research procedure"

Given this trial of dietary protein that are approved for use in Australia with no additional information being collected other than that collected as per usual care the Victorian investigators have answered Victorian Specific Module question number 1.1 (ii) as no.

Given collection of the information is without the consent of individuals or their medical treatment decision maker the Victorian investigators have answered Victorian Specific Module questions 2.1 through to 2.3 and 2.7

As no blood or tissue is being obtained Victorian investigators have left all questions in the Victorian Specific Module section 3 blank.

10.5 Confidentiality of patient data

All patients admitted to the participating site ICU and receiving study EN will be enrolled and will be allocated a unique study number. The site research coordinator will compile an enrolment log including the patient's name, date of birth, hospital identification number and unique study number and date and time of randomisation. Subsequent data will be identified by the unique study number only. The enrolment log and study data will be kept separately. The data will be kept confidentially at the site according to each sites' identifiable data storage policy such as, a locked cabinet in the research office. For further information see section 8 "Data Management".

11. TIMELINE

Timeframe Indicator:	Milestone:
May-July 2021	Protocol development
August 2021	Ethics submission
August 2021	Sites Clinical Trial Research Agreement
September 2021	Research governance submission
October 2021 - March 2022	Finalise study documents
May-August 2022	Commence recruitment (staggered start)
May- August 2022	Patient recruitment completed (staggered finish)

12. FUNDING

Funding to support site start-ups and data collection will be provided from the National Health and Medical Research Council Project Grant (APP1144496), Protein Absorption and Kinetics in Critical Illness, PI: Marianne Chapman. Additional funding is currently being sourced for this study to support further site payments.

13. TRIAL REGISTRATION

The study was prospectively registered with the Australian New Zealand Clinical Trials Registry (http://www.anzctr.org.au). Trial ID: ANZCTR12621001484831.

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15. SUPPLEMENTARY FILES

1. Appendix 1

Nutrison Protein Intense product information

2. Appendix 2

Nutrison Protein Plus product information

3. Appendix 3

Australian and New Zealand Intensive Care Society Adult Patient Database minimum data set

4. Sub-study

The effect of augmented PROtein administration on computed tomography (CT) - derived Skeletal muscle in Critically ill Adults receiving enteral Nutrition: A sub-study to TARGET Protein cluster randomised trial (PRO-SCAN Study)

BACKGROUND

Skeletal muscle plays an important role in movement, but also has vital immune and metabolic functions during illness [35]. Intensive care unit (ICU) acquired weakness is a significant complication of critical illness [36], with observational studies reporting a 18% decrease in quadriceps skeletal muscle mass in the first 10 days of ICU admission [15]. Muscle wasting has been associated with increased length of mechanical ventilation and increased

ICU and hospital mortality [37, 38] with survivors of critical illness heavily impacted by prolonged functional disability and reduced quality of life [39].

Observational studies have shown that 60-70% of patients admitted to ICU have lower than normal muscle mass, with associated increased mortality [40, 41]. Poor muscle quality, in addition to muscle mass, is an important determinant of functional outcomes following critical illness [42], with increased intramuscular lipid content (myosteatosis) and fibrous tissue a major cause of poor muscle quality in the critically ill [43]. The pathogenesis of muscle wasting in critical illness is multi-factorial with suboptimal nutritional delivery, particularly dietary protein, a potential contributor [44]. Increased protein administration has been reported to attenuate muscle wasting in a number of small randomised controlled trials in critically ill patients [6, 45]. It is therefore plausible that augmented protein administration, to that recommenced in international nutrition guidelines, has the potential to reduce muscle wasting in critical illness.

Reference methods to assess muscle mass and quality, such as dual-energy x-ray absorptiometry, are not logistically feasible for widespread use in the ICU setting [46]. Computed tomography (CT) scans can be used to quantify skeletal muscle cross-sectional area (CSA) at the third lumbar (L3) region as an evaluation of muscle mass, which has been shown to correlate with whole body muscle mass in a number of clinical conditions [46-48]. In addition, these CT images can be used to identify myosteatosis within muscle fibres to evaluate skeletal muscle density as a measure of muscle quality [42, 48]. These techniques have been recently used to assess skeletal muscle mass and quality in the critically ill [49]. Critically ill patients frequently have CT scans performed for clinical purposes, and for some patients these images are obtained repeatedly throughout their ICU admission. Therefore, these images can be repurposed and analysed for research purposes to advance our understanding without additional risk or burden to patients [50, 51].

TARGET Protein is a large binational, multicentre, cluster-randomised trial comparing augmented protein doses recommended in international guidelines to current usual care. **This sub-study proposes to utilise the images obtained as part of clinical care to evaluate skeletal muscle mass and quality to improve our understanding of the results from the main trial**.

AIMS

The co-primary aims of this sub-study of TARGET Protein are to evaluate whether: (1) augmented administration of enteral protein attenuates skeletal muscle loss; (2) baseline muscle mass affects clinical outcome in response to augmented administration of enteral protein. Secondary aims include determining the effect of augmented enteral protein on skeletal muscle quality; and to determine the effect of greater exposure to augmented enteral protein on skeletal muscle quantity and quality.

HYPOTHESES

1. Augmenting dietary protein to critically ill adults when compared to usual care attenuates loss of skeletal muscle mass as determined by CT-derived L3 skeletal muscle CSA

2. Critically ill adults admitted with lower than normal muscle mass, as determined by CT-derived L3 skeletal muscle CSA have improved clinical outcomes following augmented protein than those with admitted with normal muscle mass.
3. Augmenting dietary protein to critically ill adults when compared to usual care attenuates loss of skeletal muscle quality as determined by CT-derived L3 skeletal muscle density.
4. Augmenting dietary protein to critically ill adults when compared to usual care attenuates loss of skeletal muscle mass as determined by CT-derived L3 skeletal muscle CSA in patients that receive greater than 7 days of intervention

OUTCOMES

Co-Primary outcome

- To assess the effect of augmented enteral protein, when compared to usual care, on change in muscle mass as determined by CT-derived L3 skeletal muscle CSA from baseline (prior to TARGET Protein enrolment) to subsequent measurements, overall and by treatment group.
- To determine the effect of lower than normal muscle mass on ICU admission, as determined by CT derived skeletal muscle CSA (defined as <170cm^2 for males and <110cm^2 for females [40]) on clinical outcomes (days alive and out of hospital at day 90 and mortality) in response to augmented protein administration.

Secondary outcome

- To assess the effect of augmented enteral protein, when compared to usual care, on change in skeletal muscle quality as determined by CT-derived L3 skeletal muscle density from baseline (prior to TARGET Protein enrolment) to subsequent CT measurements, overall and by treatment group.
- To assess the effect of greater than 7 days of augmented dietary protein, when compared to usual care, on attenuation skeletal muscle mass loss as determined by CT-derived L3 skeletal muscle CSA.

RESEARCH PLAN

Design

A sub-study of a cluster randomised, cross-sectional, double cross-over, registry-embedded, pragmatic clinical trial evaluating the effect of augmented administration of enteral protein to critically ill adults.

This sub-study will address two separate but complementary research questions to the aim of the parent trial (TARGET Protein) and contributes to the overarching objectives by using a subset of study participant's images that were obtained for routine care but would be otherwise unreported.

Study Participants

All patients enrolled into TARGET Protein at participating sites will be eligible if they had a CT scan that meets quality checks, capturing a transverse slice at the level of the third lumbar (L3) vertebra as part of routine clinical care at any point from 72 hours before ICU admission until 48 hours after TARGET Protein study EN was commenced.

Study Procedures

All CT scans of the chest, abdomen or pelvis performed for clinical care will be identified from electronic medical records in patients enrolled into TARGET Protein at participating sites. Subsequent CT scans are required for the analysis of only one of the dual primary outcomes (i.e., aim 1). For a subsequent CT scan to be included, the scan must be taken a minimum of 5 days after the initial scan to ensure clinically meaningful effect of nutrition intervention. The appropriate CT slice at L3 will be identified and images extracted.

The digital imaging and communications in medicine (DICOM) data files will be deidentified at the site level, labelled with date of scan and study ID, and transferred to the coordinating centre via an encrypted file. At the coordinating centre files will be uploaded into Voronoi Health Analytics Data Analysis Facilitation Suite software for analysis. The skeletal muscle will be identified using boundaries in Hounsfield units (HU) [49]. Skeletal muscle CSA in cm^2 will be computed by summing the skeletal muscle tissue pixels and multiplying by the surface area of each pixel [49]. This methodology has been previously reported in critically ill patient[52]. Skeletal muscle density will be assessed by calculating the mean radiologic muscle attenuation of all muscles visible at the L3 level measured in HU [49].

DATA MANAGEMENT

Data Collection

CT images will be extracted from electronic medical records and de-identified at participating sites. The following baseline and clinical data will be extracted from the TARGET Protein database: patient demographics (age, sex, weight), admission diagnosis, duration of ICU and hospital stay, hospital outcome, clinical frailty score on admission, ventilation status and duration of mechanical ventilation. Daily nutrition data collected on days 1-5, 10, 20, 30 and 90 while receiving TARGET Protein study EN will also be extracted including the volume of TARGET Protein study EN delivered, delivery of parenteral nutrition and protein supplements, and date and time of commencement and cessation of TARGET Protein study EN.

Data Analysis

Baseline skeletal muscle area, and the association with clinical outcomes will be assessed both as a continuous covariate and, depending on the observed relationships, as a binary grouped variable (low/normal). For the binary clinical outcome of hospital death, interaction effects for study group with age, gender and the presence of sepsis will be explored. For clinical outcomes where death is a competing event, (e.g. duration of invasive mechanical ventilation, ICU and hospital LOS), competing risks regression will be employed using a cumulative incidence framework as per Fine and Grey [53]. Hospital/ICU site will be included as a random effect if study numbers allow and confidence intervals will be calculated using robust standard errors for all models. As all analyses will be regarded as exploratory, no adjustment will be made for multiple comparisons and p-values provided for perspective only, rather than dichotomous statistical significance [54]. All analyses will be undertaken using Stata/MP 17.0 (StataCorp LP, Texas, USA).

This sub-study relies on the number of CT images obtained for clinical care; hence, no sample size calculations have been performed; however, it is anticipated that TARGET Protein will recruit over 3,000 patients from 8 participating sites. Based on a previous research study utilising CT scans collected for clinical care, conducted at one of the participating TARGET Protein sites [50], approximately 11 patients per year had at least one abdominal CT scan, however this may be an underestimate as the study compared CT scans to quadriceps ultrasound (collected from a separate research study) and therefore limited the sample size, as inclusion criteria for this study required the completion of both CT scans and quadriceps ultrasound. A modest estimate of 20 patients per year at each site would have at least one CT scan, therefore a total of 160 patients are expected to be included (80 patients in each group) from all 8 sites. Furthermore, based on a retrospective single centre study [17] it is estimated that 20% of these patients will have at least two CT scans and 5% greater than two scans, it is therefore anticipated that over 200 CT scans will be included in this sub-study.

Data Category

Re-identifiable: A master enrolment log of participants will be completed for the sub-study at each participating site. This will record identifying details of each participant and will be used to assign each participant a sub-study number. The enrolment logs will be retained securely and confidentially at each participating site. The investigators propose that the data obtained remain re-identifiable to enable the data to be interrogated/corrected in the case of unexpected aberrant data e.g. database and data entry errors. Re-identifiable data may be used for data verification and query resolution and only referred to, if required, by local site investigators.

Data Storage and Transfer

The local site principal investigators will be responsible for data storage at the participating sites. CT digital files will be extracted locally and all patient identifying labels removed. The anonymised files will be transferred to the coordinating centre via an encrypted file. Sites will store enrolment logs and non-identifiable (anonymised) participant data according to local site data management policies.

Data Access

The principal investigator is responsible for the data overall. Each participating centre retains ownership of their local site data managed by the site principal investigator. Only those persons directly involved in data collection and analysis will have access to site study data. The pooled anonymised data will be held at the coordinating centre and remain the property of the TARGET Protein Management Committee and only study investigators will have access to stored non-identifiable pooled electronic data.

Data Retention and Destruction

Both electronic and hard copy study data will be retained for a period of fifteen years following completion of the study. Following that time, information stored will be disposed of in a confidential manner as per site policy. In addition to the processes described above, the investigators acknowledge that data may otherwise be discoverable through processes of law or for assessing compliance with research procedures.

Ethical Consideration

The TARGET Protein study has been approved by the Central Adelaide Local Health Network Human Research Ethics Committee (CALHN HREC), (HREC reference number: 2021/HRE00248) for a full waiver of consent.

This sub-study involves the use of data that is already collected for clinical care and will have no effect on patient outcomes; Moreover, the sub-study addresses separate but complementary research questions to the aim of the parent (TARGET Protein) trial and contributes to the overarching objectives of TARGET Protein by using a subset of study participant's images that were obtained for routine care but would be otherwise unreported. Therefore, we propose this sub-study continues to align with the approved full waiver of consent the CALHN HREC has granted.

Sub-study Sites and Authorship

All sites participating in TARGET Protein will be invited to participate in this sub-study. Sites that provide 10 or more paired images will be invited to have three contributors from their site listed as authors. The site PI will put forward the names of these co-authors who will be approved by the sub-study PI, Mr Matthew Summers. The listed co-authors will be followed by on behalf of the TARGET Protein investigators.